

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 7: Load & Trim Sheet Execution & STAB TRIM Setting

Syllabus Source: CAE Oxford (pp. 125-144) | Pass Target: $\geq 90\%$ AviationExam

1. THE ARCHITECTURE OF THE LOAD AND TRIM SHEET

The Load and Trim Sheet is the official legal document required before every commercial departure. It fulfills two vital safety functions:

- **Mass Check (The Upper Section):** Verifies that the aircraft does not exceed any of its 3 structural mass limits: Maximum Zero Fuel Mass (MZFM), Maximum Take-Off Mass (MTOM), and Maximum Landing Mass (MLM).
- **Balance & Trim Check (The Lower Section):** Graphically plots the Center of Gravity using an 'Index Grid' (trim drop lines) to verify that the CG is inside the flight envelope and to read the Take-Off Stabilizer Trim setting (STAB TRIM in units ANU / AND).

2. THE INDEX SYSTEM & REDUCTION FORMULA

Because raw moments on transport jets run into millions of kilogram-meters, calculations are simplified into small manageable 'Index Units':

- **The Index Formula:** $\text{Index} = ((\text{Mass} \times (\text{Arm} - \text{Reference Datum})) / \text{Divisor (C)}) + \text{Constant (K)}$
- **Divisor (C):** Typically 10,000 or 100,000. Reduces the large moment into a small 2 or 3 digit number.
- **Constant (K):** A positive baseline offset (e.g. +50 or +100) added so that all calculated indices remain strictly positive numbers, eliminating negative calculation errors!

3. TAKE-OFF STABILIZER TRIM (STAB TRIM SETTING)

Before take-off, the trimmable horizontal stabilizer (THS) must be set to a specific pitch trim angle corresponding to the Take-Off % MAC:

- **Aft CG -> Low Nose-Up Trim:** At an aft CG, the aircraft naturally pitches up easily. STAB TRIM is set to a low nose-up value (e.g. 3.0 units ANU).
- **Forward CG -> High Nose-Up Trim:** At a forward CG, heavy nose-down moments require a large nose-up stabilizer setting (e.g. 7.5 units ANU) to allow rotation at V_r without excessive pilot pull force!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Last Minute Change (LMC) Protocol

Scenario: Late passenger boards with baggage (Adult male 84 kg + bag 13 kg = 97 kg). Step 1: Check LMC mass does not exceed company tolerance limit (typically 500 kg). Step 2: Add 97 kg to Zero Fuel Mass, Take-Off Mass, and Landing Mass. Step 3: Verify all three updated masses remain below structural and performance limits: - ZFM + 97 kg $\leq$ MZFM - TOM + 97 kg $\leq$ Regulated MTOM - LM + 97 kg $\leq$ MLM Step 4: If seated near CG, pitch trim change is negligible; if in aft row, adjust STAB TRIM units per LMC table.

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Index & Reading Trim Sheet

SCENARIO (Standard EASA Exam Trim Sheet Problem): An airline uses the following Index

Reduction Formula for its twin-jet: $\text{Index} = [\text{Mass (kg)} \times (\text{Arm (m)} - 15.0 \text{ m}) / 1,000]$

+ 50.0 Given the following loading changes to the Dry Operating Index: - Dry Operating

Mass (DOM): 32,000 kg with an Initial Index of 52.0 units - Cargo loaded in

Compartment 1: 2,000 kg at Arm = 8.0 meters - Cargo loaded in Compartment 4: 1,500 kg

at Arm = 22.0 meters QUESTION: Calculate the New Zero Fuel Mass (ZFM) and the New Zero

Fuel Index! SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate New Zero Fuel

Mass: - $\text{ZFM} = \text{DOM} + \text{Compartment 1} + \text{Compartment 4}$ - $\text{ZFM} = 32,000 \text{ kg} + 2,000 \text{ kg} +$

1,500 kg = 35,500 kg. Step 2: Calculate Delta Index for Compartment 1 (Arm = 8.0 m):

- $\text{Delta Index}_1 = [\text{Mass} \times (\text{Arm} - 15.0)] / 1,000$ - $\text{Delta Index}_1 = [2,000 \times (8.0 -$

15.0)] / 1,000 = $[2,000 \times (-7.0)] / 1,000 = -14.0$ units! - (Notice it is negative

because the arm is forward of the 15.0 m reference!). Step 3: Calculate Delta Index

for Compartment 4 (Arm = 22.0 m): - $\text{Delta Index}_4 = [\text{Mass} \times (\text{Arm} - 15.0)] / 1,000$

- $\text{Delta Index}_4 = [1,500 \times (22.0 - 15.0)] / 1,000 = [1,500 \times (+7.0)] / 1,000 = +10.5$

units! Step 4: Sum the Initial Index with both Delta Indices to find New ZFM Index:

- $\text{New ZFM Index} = \text{Initial Index} + \text{Delta Index}_1 + \text{Delta Index}_4$ - $\text{New ZFM Index} =$

52.0 + (-14.0) + (+10.5) = 48.5 units! FINAL ANSWER: New ZFM = 35,500 kg; New ZFM

Index = 48.5 units. On the trim grid, enter at Index 48.5 to read % MAC.

[!] EXAM TRAP (AVIATIONEXAM) - STAB TRIM Setting Units Trap (ANU vs AND)

- ANU = Aircraft Nose Up (positive nose-up trim). - AND = Aircraft Nose Down. -

Forward CG ALWAYS requires MORE ANU (higher nose-up stabilizer trim) to assist elevator rotation at Vr!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the primary purpose of using 'INDEX UNITS' instead of raw moments in aircraft load and trim sheet

- [A]: To convert metric units to imperial units
- [B]: To reduce huge moment numbers (millions of kg.m) into manageable small numbers, preventing human calculation errors in time-critical airline turnarounds
- [C]: To calculate cruise speed
- [D]: To record passenger names

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Moments for transport airliners are enormous numbers. Dividing by a constant divisor (e.g. 10,000) and adding a base constant produces compact index values (e.g. 40-60) easily plotted on graphical grids.

Q2: When setting the Take-Off Stabilizer Trim (STAB TRIM) on a commercial transport jet, what setting is required

- [A]: Zero trim
- [B]: A GREATER Aircraft Nose Up (ANU) trim setting, to counter heavy nose-down moments and ensure adequate elevator rotation authority at Vr
- [C]: Full Aircraft Nose Down (AND) trim
- [D]: Trim does not depend on CG

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A forward CG produces a large nose-down moment. A higher ANU (nose-up) stabilizer setting is mandatory so the pilot can rotate the nose at V_r without exceeding control column pull limits.

Q3: On a load and trim sheet, if an aircraft's Take-Off Mass is plotted and falls OUTSIDE the certified envelope

- [A]: Increase engine thrust to TOGA
- [B]: Departure is STRICTLY PROHIBITED until payload or fuel is redistributed or offloaded to bring the point within the certified envelope
- [C]: Notify ATC on climb out
- [D]: Fly at a lower altitude

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Flying outside the M&B envelope is illegal and causes loss of control or structural overload. The flight crew must re-trim by shifting cargo, moving passengers, or offloading fuel before departure.